

SOC-5811 Week 15: What have we learned?

Nick Graetz

University of Minnesota, Department of Sociology

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REVIEW: BIG PICTURE

What is the point of building a model?

- **Prediction**



REVIEW: BIG PICTURE

What is the point of building a model?

- ▶ **Prediction**
- ▶ **Comparison**



REVIEW: BIG PICTURE

- ▶ **Regression is a mathematical tool for making predictions.**



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- ▶ Regression coefficients can always be interpreted as average comparisons of predictions.

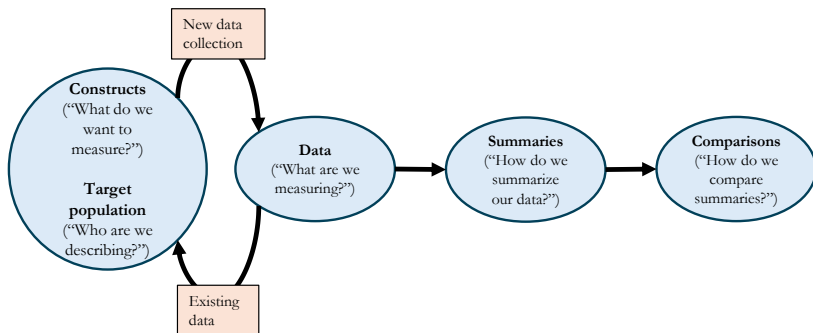


REVIEW: BIG PICTURE

- ▶ **Regression is a mathematical tool for making predictions.**
- ▶ Regression coefficients can always be interpreted as average comparisons of predictions.
- ▶ Regression coefficients can sometimes be interpreted as effects.



REVIEW: BIG PICTURE



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- ▶ Generalization:



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- ▶ Generalization:
 - ▶ **Constructs:** Do our measures generalize to concepts?
 - ▶ **Target population:** Does our sample generalize to the population?
 - ▶ **Forecasting:** Does our model of the present generalize to the future?
 - ▶ **Causal inference:** Does our comparison generalize to a causal effect?



REVIEW: BIG PICTURE

What do we need in order to interpret our comparison as a causal effect?

- ▶ **Internal validity** (e.g., exchangeability)



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What do we need in order to interpret our comparison as a causal effect?

- ▶ **Internal validity** (e.g., exchangeability)
- ▶ **External validity** (e.g., transportability)



REVIEW: MODELS

What does it mean to “build” a model?



REVIEW: MODELS

- ▶ **Estimand:** What is my goal? What is my target comparison I want to infer something about? Example: I want to know if income varies by education, conditional on age.



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- ▶ **Estimator:** How do I relate my population estimand to real sample data in a way that let's me identify its *most likely* value given that sample data?



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- ▶ **Estimand:** What is my goal? What is my target comparison I want to infer something about? Example: I want to know if income varies by education, conditional on age.
- ▶ **Estimator:** How do I relate my population estimand to real sample data in a way that let's me identify its *most likely* value given that sample data?
- ▶ **Estimation:** How do I optimize my estimator (i.e., calculate the most likely value of my estimand)?



REVIEW: MODELS

1. **Estimand:** Choose a functional form for the model and highlight the target parameter (“estimand”).

$$y_i = \beta_0 + \boxed{\beta_1} x_i + \epsilon_i$$

2. **Estimator:** Choose an estimator that relates the model to real observed data (e.g., OLS) and is unbiased, consistent, and efficient.

$$\text{Sum of squared residuals} = \sum_{i=1}^n (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$$

3. **Estimation:** Estimate parameters by optimizing the estimator; sometimes this can be done analytically, sometimes it requires brute computational force.

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REVIEW: STATISTICAL INFERENCE

- ▶ Our models consist of a set of **parameters**: the unknown numbers (random variables) that determine a statistical model.
- ▶ We often want to do **statistical inference** on these parameters because they represent comparisons that map to our research questions.
- ▶ This involves **uncertainty**.



REVIEW: STATISTICAL INFERENCE

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- ▶ How do we estimate the standard error?
 - ▶ Analytic approach: formulas.



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 - ▶ The standard error depends on sample variability and sample size.
- ▶ How do we estimate the standard error?
 - ▶ Analytic approach: formulas.
 - ▶ Simulation approach: bootstrapping.

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- ▶ For inference, we use the **p-value**: the probability that the null model would produce a parameter estimate as extreme as your observed estimate if the null model is correct.
- ▶ Our convention is that a p-value of less than 0.05 represents “statistical significance.”



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 - ▶ We want to isolate a more specific comparison (e.g., age standardization).
 - ▶ We want to estimate a causal effect, so we control for confounders.

REVIEW: CAUSAL INFERENCE

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- ▶ The fundamental problem of causal inference.
 - ▶ We can only ever see one outcome at one time for any given unit, even though they have multiple *potential outcomes*.
- ▶ We can attempt to estimate **average treatment effects** by predicting potential outcomes.

REVIEW: CAUSAL INFERENCE

- ▶ Causal inference is a prediction problem: we must predict potential outcomes using average observed outcomes from exchangeable units.



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- ▶ Causal inference is a prediction problem: we must predict potential outcomes using average observed outcomes from exchangeable units.
- ▶ The regression model is one way of generating the predictions needed to estimate the ATE.



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- ▶ **Confounder:** any variable that affect both the likelihood of treatment and likelihood of outcome.
- ▶ **Mediator:** any variable on the causal pathway between treatment and outcome.
- ▶ **Moderator:** any variable that affects the relationship between treatment and outcome.



REVIEW: PANEL DATA

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- ▶ Panel data: A data set where we observe the same units (e.g., individuals, firms, countries, schools) over more than one time period.
- ▶ New estimator: fixed effects.
- ▶ Fixed effects for the unit of analysis removes time-invariant confounding.
- ▶ Do I care about within- or between-unit variation for my research question?

REVIEW: ASSUMPTIONS

Key assumptions:

- ▶ Validity
- ▶ Representativeness
- ▶ **Additivity and linearity**
- ▶ Independence of errors
- ▶ Equal variance of errors
- ▶ Normality of errors

REVIEW: ASSUMPTIONS

- ▶ Linear transformations



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- ▶ Linear transformations
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- ▶ Linear transformations
 - ▶ Shifting (i.e., add/subtract a constant)
 - ▶ Rescaling (i.e., multiply/divide by a constant)



REVIEW: ASSUMPTIONS

- ▶ Non-linear transformations



REVIEW: ASSUMPTIONS

- ▶ Non-linear transformations
 - ▶ Logs

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REVIEW: ASSUMPTIONS

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- ▶ Examining heterogeneous treatment effects (transform/interact the treatment variable).

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When do we think about **transforming** continuous variables or including **interactions**?

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- ▶ Try everything and use cross-validation to see if it improves out-of-sample fit.

- ▶ **Causal model**

- ▶ Examining heterogeneous treatment effects (transform/interact the treatment variable).
- ▶ Relaxing assumptions about confounding pathways (transform/interact the confounder variables).

APPLIED QUANTITATIVE RESEARCH

- Steps for an applied quantitative research project



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- ▶ **(1)** Read the literature. What do we know? What do we not know? Draw a conceptual model to guide your research question.



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- ▶ (2) Familiarize yourself with the data. Can we use existing data? Do we need new data?
- ▶ (3) Refine conceptual model into a DAG (this may include drawing the “causal field”) that is specific about measurement of *constructs* and definition of *target population*.



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- ▶ (2) Familiarize yourself with the data. Can we use existing data? Do we need new data?
- ▶ (3) Refine conceptual model into a DAG (this may include drawing the “causal field”) that is specific about measurement of *constructs* and definition of *target population*.
- ▶ (4) Specify the *estimand* that answers your research question. In other words, choose a functional form for the model and highlight your target parameter (e.g., β_1).

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APPLIED QUANTITATIVE RESEARCH

- ▶ (5) Load data into R. Make descriptive summary statistics and plots. Revisit Steps 1-4 as needed.



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APPLIED QUANTITATIVE RESEARCH

- ▶ **(5)** Load data into R. Make descriptive summary statistics and plots. Revisit Steps 1-4 as needed.
- ▶ **(6)** Choose estimator (e.g., OLS) and fit model.
- ▶ **(7)** Test for non-linearity and non-additivity in the relationship between treatment and outcome.



REVIEW

- ▶ Wednesday (12/10): What have we not learned?



REVIEW

- ▶ Wednesday (12/10): What have we not learned?
- ▶ Please complete the Student Rating of Teaching on the menu on the left of Canvas (due 12/10).

